Actuator and Contact Reuse in an Articulated Arc–Leg Hexapod

Abstract— **Wheels transport efficiently on level ground, whereas articulated legs place contacts across discontinuities; conventional wheel–leg hybrids commonly retain both mechanisms and their associated actuation. We study a different compromise: each of six 3-DoF legs terminates in one actuated, open C-shaped arc that is reused as a planted foot, a rolling contact, and a candidate stair-edge hook. One controller runs an alternating-tripod reference and continuous arc rotation on the same six distal joints: the distal rolling rate is superposed on the differentiated walking reference, requiring neither a dedicated wheel actuator nor a mechanical transformation. In a 500-Hz MuJoCo development model, an as-implemented controller comparison at a commanded $0.18\,\mathrm{m\,s^{-1}}$ yielded $0.117\,\mathrm{m\,s^{-1}}$ for articulated walking, $0.090\,\mathrm{m\,s^{-1}}$ for distal-only rotation, and $0.200\,\mathrm{m\,s^{-1}}$ for coordinated arc locomotion. Distal-only rotation reached the upper landing of a three-step $0.15\,\mathrm{m}$-riser profile, while only the coordinated controller reached the landing of the $0.20\,\mathrm{m}$-riser, $0.35\,\mathrm{m}$-tread profile. A force–torque balance shows why a radius-only wheel-height limit is insufficient for an actuated distal joint; a matched closed-wheel control is therefore required to attribute the $0.15\,\mathrm{m}$ outcome to the opening. Walk–roll and roll–walk control transitions completed in 0.342 and $0.660\,\mathrm{s}$. These deterministic trials define hypotheses and a reproducible validation protocol, not physical performance or statistical superiority.**

I. INTRODUCTION

Level-ground speed and discontinuous-terrain mobility reward different contact mechanics. A wheel preserves a rolling constraint and avoids repeated foot placement; an articulated leg can place a contact and move the body relative to that contact. Hybrid robots often carry both a leg and a driven wheel [1], [2]. This increases capability, but the wheel remains a dedicated subsystem. At the opposite extreme, rotating-leg robots obtain impressive intrinsic mobility from few actuators [3]–[5], but their contact path and body motion are strongly coupled.

We ask a narrower mechanical question: *can the last actuated link of a fully articulated hexapod be shaped so that the same joint and contact surface support foot placement, rolling transport, and stair-edge engagement?* The resulting platform has 18 actuators—three per leg—and an open, C-shaped TPU arc as each distal link (Fig. 1). No dedicated wheel is deployed and no hardware transforms between regimes. Instead, the contact interpretation changes: the lowest arc patch is held stationary during walking, the patch advances around the arc during rolling, and an arc boundary or tread patch reacts against a stair nosing during ascent.

This paper contributes: 1) a multifunctional arc–leg morphology with an explicit geometric model of its rolling and stair-edge contacts, including a force–torque balance that distinguishes a passive horizontal push from an actuated wheel; 2) Arc-Reuse Gait, a controller that runs the tripod walking reference and continuous arc rotation *simultaneously* on the same six distal joints by superposing a velocity-mode rolling rate on the differentiated walking reference; and 3) a same-model development benchmark—articulation only, distal rotation only, and the combined controller—together with a matched-parameter, matched-randomness validation protocol that can test the morphology separately from controller tuning.

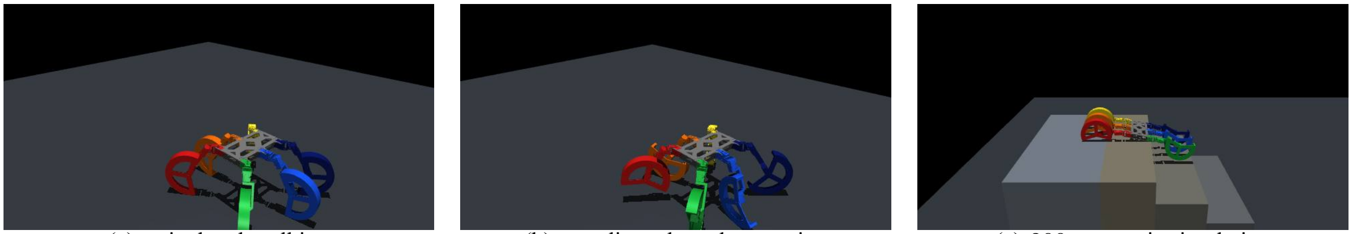
All numbers in the present draft come from a deterministic MuJoCo model; the nominal flat, stair, and transition conditions are single trials and the uneven-terrain condition uses three seeds. They describe the current model and motivate the controlled experiments of Sec. VI; they are not confidence intervals or physical-performance claims.

II. RELATED WORK AND PROBLEM FORMULATION

A. Rotating Legs and Wheeled Legs

RHex showed that one continuously rotating compliant leg per hip can combine mechanical simplicity with rugged mobility under a clock-driven tripod gait [3], [6]; a dedicated controller later demonstrated repeated ascent of full-size stairs [7]. The X-RHex family carried that morphology into a research platform [8]. Whegs abstracted compliant, multi-spoke appendages and coupled propulsion to reduce actuator count [4]; Q-Whex later used quasi-wheels and reported stair climbing with geometry-dependent phase patterns [5]. The motion and power demand of C-legged hexapods have also been modeled and checked on a physical test bench [9]. The half-circle RHex leg is the closest prior morphology to ours and the distinction must be stated precisely: an RHex leg is a compliant C driven by a single hip rotation, so its contact point and the body pose are determined by the same input. Our C is the third joint of a 3-DoF leg, so the leg can hold the arc still, place it, or spin it while the body is posed independently, and it is this separation—not the C shape—that the ablation of Sec. VI tests.

A second family transforms the mechanism itself. Wheel Transformer deploys passive spokes when a wheel meets an obstacle [10], while Quattroped and TurboQuad change a leg into a wheel with a dedicated transformation degree of freedom [11], [12]. These recover both regimes at the cost of a transformation mechanism and its actuation. A third family keeps a wheel drive alongside a full leg: ASTERISK H switches a hexapod continuously between rolling and tripod motion using sensed obstacles [13], while Cassino Hexapod III places omni-wheels at anthropomorphic leg tips [1], ANYmal on Wheels integrates wheel constraints into whole-body optimization [2], [14], [15], and two-wheeled jumpers



(a) articulated walking (b) coordinated arc locomotion (c) 200-mm stair simulation

Fig. 1. One actuated C-shaped distal link is reused in three contact regimes. Images are reproducible MuJoCo captures from the benchmark conditions reported in Sec. VI; no physical-performance claim is inferred from the renderings.

exploit the same separation in a smaller footprint [16]. They demonstrate the value of simultaneous wheel and leg control, but retain a wheel drive in addition to the leg chain. Our design does not claim that hybrid locomotion is new. The distinction tested here is *actuator and contact reuse*: the distal leg joint itself produces the foot trajectory, the rolling motion, and the edge engagement of one incomplete arc, with no transformation state and no additional drive.

B. Operating-Envelope Problem

Let the body command be

$$\vec{c} = [v_x,\ v_y,\ \omega_z]^\top \in \mathcal{C}, \qquad m \in \{W, R, C\}, \tag{1}$$

for walking, rolling, and climbing. Let $\vec{q} \in \mathbb{R}^{18}$ be the actuated joint vector and Γ_i the active contact subset of distal arc i. The design seeks one morphology $\mathcal{M}$ and mode-dependent control laws π_m rather than a transformed mechanism $\mathcal{M}_m$:

$$\vec{q}^{\star}_k = \pi_m(\vec{x}_k, \vec{c}_k; \mathcal{M}), \quad \mathcal{M}_W = \mathcal{M}_R = \mathcal{M}_C. \tag{2}$$

The relevant outcome is an operating envelope, not a single maximum speed,

$$\mathcal{O} = \{\bar{v}_x, e_v, C_{\text{mt}}, D_{\text{slip}}, P_{\text{top}}(h,d), t_{\text{top}}, t_{\text{switch}}\}, \tag{3}$$

where h, d are riser and tread dimensions. We test three hypotheses. H1: coordinated rolling reaches a higher mean forward speed than either walking or distal-only rotation on the same model. H1 is deliberately stated as speed and not as command tracking, because a controller that overshoots a fixed command scores worse on e_v while travelling faster; both quantities are reported separately in Sec. VI. H2: on the open-arc model, coordinated articulation reaches a stair profile on which distal-only rotation fails. A claim about the opening itself requires the matched closed-wheel ablation defined in Sec. V. H3: mode switching does not violate a $60°$ tilt bound.

We also define the qualitative column of Table I. Let J_i map leg i's joint rates to the pair (tangential advance of the contact patch along the distal surface, position of that contact relative to the body). Authority is *coupled* when rank $J_i = 1$, so that one input fixes both; *separated* when a dedicated drive makes the two independent at the cost of an extra actuator; and *partly separated* when, as here, three leg joints can choose the pair over a bounded set but the distal joint is shared between them.

TABLE I

Morphological position of the proposed platform

Platform	leg actuation	rolling element	extra wheel drive	body/contact authority
RHex [3]	1/leg	compliant arc	no	coupled
Whegs I [4]	coupled	3-spoke wheg	shared	coupled
Q-Whex [5]	1/leg	quasi-wheel	no	coupled
Cassino III [1]	articulated	omni-wheel	yes	separated
Proposed	3/leg	open distal arc	no	partly separated

TABLE II

Actuation, contact and distal geometry used in simulation

Quantity	Value	Role
IDs 1–6	MX-28AT, 77 g	proximal
IDs 7–12	XM430-W350-T, 82 g	middle
IDs 13–18	XM430-W210-T, 82 g	distal arc
FR07 / FR12	20/46 g	frames
Arc radii r_i/r_o	112.5/122.5 mm	inner/outer
Arc width / opening ψ	44 mm / 135°	TPU contact
Sliding friction μ	1.0	arc–ground
Contact softness	3 mm, $\zeta = 1$	`solimp/solref`
Modeled mass	4.161 kg	MuJoCo only
Physics / control	500/50 Hz	integrate / target

III. PLATFORM AND CONTACT MODEL

A. Mechanical and Simulation Model

The six legs are numbered front to rear as right $1, 3, 5$ and left $2, 4, 6$. Actuator IDs 1–6 rotate the proximal links, IDs 7–12 the middle links, and IDs 13–18 the distal arcs. The MuJoCo model has 57 bodies, one floating base and 18 revolute joints, and a total modeled mass of 4.161 kg. Its component masses encode measured aluminum frames and actuators plus estimated 15%-infill polymer parts; therefore the total is a simulation parameter until the reassembled robot is weighed. Table II distinguishes that model quantity from verified part data.

Each actuator is a voltage-input DC motor with position feedback rather than an ideal torque source. For motor constants K, R, the outer PD loop and voltage saturation are

$$\tau_i^c = k_{p,i}(q_i^\star - q_i) + k_{d,i}(\dot{q}_i^\star - \dot{q}_i), \tag{4}$$

$$V_i = \mathrm{sat}_{[-V_n, V_n]}\left(\frac{R_i}{K_i}\tau_i^c + K_i\dot{q}_i\right). \tag{5}$$

Each group is specified by its rated triple (voltage, stall torque, no-load speed), from which $K = V_n/\dot{q}_{\text{nl}}$ and $R = KV_n/\tau_{\text{stall}}$ follow: $(12\,\text{V}, 2.5\,\text{N m}, 5.76\,\text{rad/s})$,

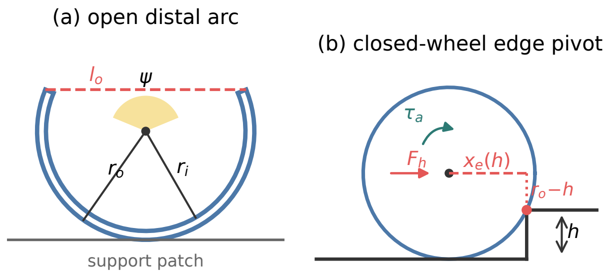


Fig. 2. Contact geometry of one distal link. (a) The open sector: outer and inner radii r_o, r_i, opening angle ψ and opening chord l_o, with the lowest patch acting as the support point during walking. (b) The closed-wheel pivot used to separate horizontal axle force F_h from applied axle torque τ_a. A force-only push becomes singular as $h \to r_o^-$, but this is not a universal height bound for an actuated wheel.

$(12, 4.1, 4.82)$, and $(12, 3.0, 8.06)$. The simulated gains (k_p, k_d) in SI units are $(5.73, 0.752)$, $(9.40, 0.792)$, and $(6.88, 0.264)$. Integral gain is zero; supply voltage and a stall-matched saturation torque bound the response.

B. Arc Contact and Stair-Edge Mechanics

Figure 2 fixes the notation. For outer radius r_o, inner radius r_i, and opening ψ, the opening chord is

$$l_o = 2r_o \sin(\psi/2) = 0.226 \text{ m}. \tag{6}$$

Contacts use a six-dimensional friction cone with sliding coefficient $\mu = 1.0$ and a 3 mm compliant layer, so the TPU arc is modeled as soft but not deformable. During walking, the support point is the centroid of mesh vertices within 0.1 mm of the lowest arc patch. During rolling, this patch advances with the distal angle.

Consider a single arc pivoting over a sharp riser of rise h with its centre initially one radius above the lower tread. The horizontal centre-to-edge offset is

$$x_e(h) = \sqrt{r_o^2 - (r_o - h)^2} = \sqrt{2r_o h - h^2}, \quad 0 \le h \le 2r_o. \tag{7}$$

If that arc carries load F_i, let $\tau_{a,i}$ be the favorable axle torque, $F_{h,i}$ a forward horizontal force applied at the arc centre, drive efficiency $\eta \le 1$, and safety factor $\gamma \ge 1$. A quasi-static necessary moment condition about the edge is

$$\eta\tau_{a,i} + F_{h,i}(r_o - h) \ge \gamma F_i x_e(h), \tag{8}$$

$$\mu_{\text{min}} = \frac{|F_t|}{F_n}. \tag{9}$$

For the restricted case $\tau_{a,i} = 0$ and $h < r_o$, this reduces to

$$F_{h,i} \ge \gamma \frac{F_i x_e(h)}{r_o - h}, \tag{10}$$

which diverges as $h \to r_o^-$. For $h \ge r_o$, a forward horizontal force no longer supplies a favorable moment, whereas the axle-torque term in (8) remains finite for $h < 2r_o$. Therefore $h = r_o$ is a force-only singularity, not a closed-wheel height limit for the actuated distal joint studied here.

The opening and articulation provide two additional contact mechanisms. If $h < l_o = 226$ mm and the approach orientation admits the edge, the nosing can enter the 135° gap and react against an arc boundary. This chord condition is necessary clearance, not a sufficient hook criterion. Articulation can also raise and reposition the whole arc. The present open-arc trials show when each controller succeeds, but only a mass-, radius-, torque-, and control-matched closed-wheel geometry can attribute the difference to the opening. That direct ablation is specified in Sec. V. A second dimensionless descriptor compares the opening chord with one step's diagonal, $l_o/\sqrt{h^2 + d^2}$, giving 0.87, 0.91, and 0.56 for the three tested profiles. We report it because it is the standard legged-wheel comparison, and note that it does not by itself order the observed outcomes.

IV. Arc-Reuse Gait: SIMULTANEOUS WALKING AND ARC ROTATION

The design goal is a single controller in which the tripod walking motion and the continuous rotation of the distal arcs run *at the same time* on the same six joints, rather than a mode switch between a walking machine and a driving machine. Section IV-A defines the bounded reference shared by the walking controllers, and Sec. IV-B defines the controller that produced the coordinated results in Sec. VI.

A. Alternating-Tripod Reference

Tripods $\mathcal{T}_A = \{1, 4, 5\}$ and $\mathcal{T}_B = \{2, 3, 6\}$ have offsets $\delta_i \in \{0, 1/2\}$. With cycle frequency f and duty factor D, the leg phase is

$$\phi_i(t) = (ft + \delta_i) \bmod 1. \tag{11}$$

At nominal foot position $\vec{p}_{i0} = [x_i, y_i, z_i]^\top$, the planar velocity induced by the commanded body twist and the resulting stance stroke are

$$\vec{u}_i = \begin{bmatrix} v_x - \omega_z y_i \\ v_y + \omega_z x_i \end{bmatrix}, \quad \vec{s}_i = D\vec{u}_i/f. \tag{12}$$

The elliptical workspace projection $\vec{s}_i \leftarrow \vec{s}_i/\max(1, \|[s_{ix}/s_x^{\max}, s_{iy}/s_y^{\max}]\|_2)$ prevents unreachable strides. Let $S(a) = 10a^3 - 15a^4 + 6a^5$ and $H(a) = 16a^2(1-a)^2$. The desired contact trajectory is

$$\vec{p}_i^\star = \vec{p}_{i0} + \begin{cases} (\frac{1}{2} - \phi_i/D)[\vec{s}_i^\top, 0]^\top, & \phi_i < D, \\ (S(a) - \frac{1}{2})[\vec{s}_i^\top, 0]^\top + h_f H(a)\vec{e}_z, & \phi_i \ge D, \end{cases} \tag{13}$$

where $a = (\phi_i - D)/(1 - D)$. Damped numerical inverse kinematics maps the six targets to the bounded 18-joint reference $\vec{q}^{\text{ref}}$. Two parameter sets of (11)–(13) are used and are reported here because they are not identical: the walking condition uses $(f, D, h_f, s_x^{\max}, s_y^{\max}) = (1.0\,\text{Hz}, 0.50, 25, 90, 70\,\text{mm})$ and the rolling conditions use $(0.8\,\text{Hz}, 0.58, 20, 55, 45\,\text{mm})$, the smaller stroke reflecting that rotation, not stride, is expected to carry most of the displacement.

B. Superposed Rotation on Velocity-Mode Distal Joints

Joints 1–12 track $\vec{q}^{\text{ref}}$ in position mode. The six distal joints are switched to velocity mode, which removes the one-turn bound, and are commanded as the sum of a continuous rolling term and the differentiated walking reference,

$$\dot{q}_{d,i} = \underbrace{\dot{\theta}_i^{\text{roll}}}_{\text{transport}} + \lambda \underbrace{\Delta q_{d,i}^{\text{ref}}/\Delta t}_{\text{walking}}, \qquad \lambda = 0.35, \tag{14}$$

which is the concrete form of "walk and rotate at once": neither term is gated off by the other. Let the normalized command activity be

$$\alpha(\vec{c}) = \text{clip}\left(\max_j \frac{|c_j|}{c_j^{\max}}, 0, 1\right), \quad \vec{c}^{\max} = [0.18, 0.12, 0.9], \tag{15}$$

and let the stance/swing rate ratio be the two-level function

$$\kappa_i(\phi_i) = \begin{cases} \kappa_{\text{st}} = 0.80, & i \in \text{stance}, \\ 1, & i \in \text{swing}. \end{cases} \tag{16}$$

With polarity $\sigma_i \in \{-1, 1\}$ and alignment $A_i \in [0, 1]$ obtained from a finite-difference contact tangent $\vec{t}_i = \partial \vec{p}_{c,i}/\partial q_{d,i}$ measured on the mesh,

$$A_i = \max\left(0, \ \hat{\vec{t}}_i^{\top} \frac{-\vec{u}_i}{\|\vec{u}_i\|}\right), \tag{17}$$

the unfiltered rolling rate and its first-order realization are

$$\omega_i^{\text{cmd}} = -\sigma_i \omega_0 \alpha(\vec{c}) A_i \kappa_i, \qquad \omega_0 = 4.20\,\text{rad/s}, \tag{18}$$

$$\lambda_r = 1 - e^{-\Delta t/\tau_r}, \qquad \dot{\theta}_i^{\text{roll}} \leftarrow \dot{\theta}_i^{\text{roll}} + \lambda_r(\omega_i^{\text{cmd}} - \dot{\theta}_i^{\text{roll}}), \tag{19}$$

with $\tau_r = 0.10\,\text{s}$. The walking term of (14) is clipped to $\pm 110°/\text{s}$ and low-pass filtered with $\tau_b = 0.04\,\text{s}$ so that an inverse kinematics branch change cannot appear as a rate spike. At start-up the three tripod-B arcs are staggered by $60°$; without this offset all six openings can face the ground simultaneously and the body drops.

The consequence that matters for Sec. VI is that (18) is an *activity-scaled* rate, not a no-slip rate. The commanded contact speed during stance is

$$v_c = r_o\, \omega_0\, \alpha\, \kappa_{\text{st}} = 0.411 \ \text{m}\,\text{s}^{-1} \quad \text{at } \alpha = 1, \tag{20}$$

so whenever the body advances more slowly than v_c the difference must appear as measured contact slip,

$$D_{\text{slip}} \approx \int_0^T \left| v_c - \|\vec{v}_{xy}(t)\| \right| dt. \tag{21}$$

Section VI uses (21) as a check rather than as a claim.

C. Stair State Sequence

The stair controller executes Walk→Drive→Climb, verifies each settled pose, braces the front middle joints, and then synchronizes the six distal phases. A geometry-conditioned brace uses $180°$, $184°$, or $195°$ over increasing riser bands. This is a deterministic supervisor that is given the stair dimensions; it performs no terrain perception and no foothold search.

V. EXPERIMENTAL METHOD

A. Reproducible Simulation Protocol

Experiments use MuJoCo [17] with 2 ms integration and 20 ms control. Every trial creates a fresh floating-base model and controller. The nominal flat test settles for 1 s and measures 6 s at $\vec{c} = [0.18, 0, 0]$. The three same-model restrictions are:

- A articulated tripod walking, distal joints in position mode with the continuous rolling term of (14) removed;
- B distal-only rotation, distal joints in velocity mode driven by (18) while the body and proximal targets are held at the staggered stance pose;
- C the full controller of Sec. IV-B, in which both terms of (14) are active simultaneously.

The restrictions therefore isolate control authority on one unchanged model. Two caveats follow from Sec. IV-A and are stated before the results rather than after them: A and C do not share the walking parameter set (1.0 Hz, $D = 0.50$, 90 mm stroke versus 0.8 Hz, $D = 0.58$, 55 mm stroke), and the rolling rate of (18) is scaled by normalized command activity rather than tuned per condition. Neither condition was separately optimized for maximum speed, so the comparison establishes the qualitative ordering of the three regimes on one model and not the best speed attainable by any of them.

To turn this development ablation into a publication comparison, we additionally define a locked validation protocol. All three restrictions use one common gait parameter set, actuator limits, initial distal phase, and paired perturbation sample. Contact geometry is crossed with control authority: the exported open arc is compared with a closed cylinder having the same radius, width, body mass, inertia, actuator, and collision parameters. For each terrain seed, every geometry–controller pair receives the same mass, friction, actuator-strength, base-pose, and gait-phase perturbation. The protocol and its random seeds are written to a manifest before execution. This matched experiment is intentionally reported as required validation rather than retrofitted into the single-run numbers below.

The stair set contains three 3-step profiles: $(h, d) = (0.10, 0.24)$, $(0.15, 0.20)$, and $(0.20, 0.35)$ m, where d is the minimum tread. Strategies are distal-only rotation, which holds every non-arc joint and spins the six arcs at one constant rate so that no articulation is available; a synchronized open-loop roller with a fixed front brace; and the full geometry-conditioned controller. Reaching the upper landing defines success. Transition trials measure the time from the first mode-change command until all guarded actuator conditions are satisfied, followed by 2 s recovery. Uneven-terrain smoke tests use three randomized seeds and 2 s windows; they check execution, not publication-grade robustness.

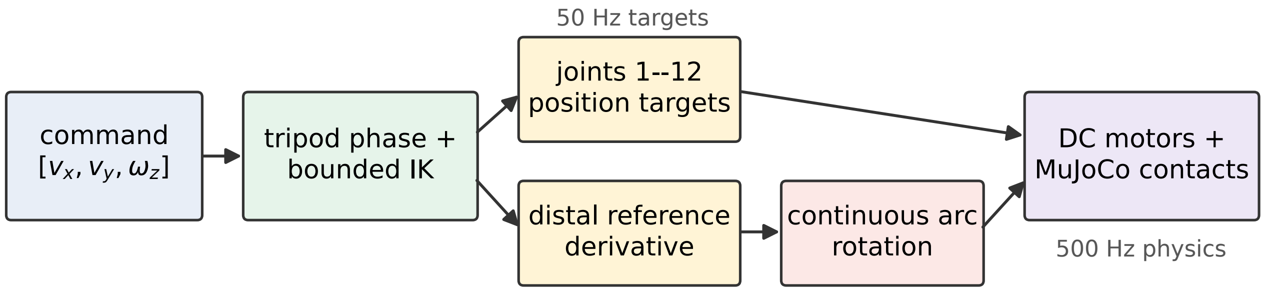


Fig. 3. Control paths. Joints 1–12 track the bounded tripod reference in position mode while joints 13–18 run in velocity mode and add continuous arc rotation to the differentiated reference, so walking and rotation are commanded at the same instant. The controller outputs 12 position targets and six velocity targets; no learned policy is used in the reported experiments.

B. Metrics

Body-frame velocity tracking error and integrated slip are

$$e_v = \sqrt{\frac{1}{T}\int_0^T \|\vec{v}_{xy}(t) - \vec{c}_{xy}\|_2^2 dt}, \tag{22}$$

$$D_{\text{slip}} = \int_0^T \frac{1}{|\mathcal{K}(t)|} \sum_{k\in\mathcal{K}(t)} \|\vec{v}^{\parallel}_{c,k}(t)\|_2 dt, \tag{23}$$

where $\mathcal{K}$ contains loaded arc–ground contacts carrying more than $1\,\mathrm{N}$, about 2.5% of the modeled body weight. Because e_v is measured against a constant command, a controller that overshoots that command is penalized; $\bar{v}_x$ and e_v are therefore read together. Absolute mechanical work and mechanical cost of transport are

$$W_{\text{abs}} = \int_0^T \sum_{j=1}^{18} |\tau_j \dot{q}_j| dt, \qquad C_{\text{mt}} = \frac{W_{\text{abs}}}{mgd_{xy}}, \tag{24}$$

with d_{xy} the net planar displacement. An electrical estimate uses $I_j = (V_j - K_j\dot{q}_j)/R_j$ and $E_e = \int \sum_j |V_j I_j| dt$. Because neither battery conversion loss nor servo electronics is modeled, E_e is diagnostic, not physical battery energy. Uprightness is $u_z = \vec{e}_z^{\top} R_{WB}\vec{e}_z$; termination occurs below $\cos 60^\circ$ or on non-finite state.

C. Scope of the Evidence

Every trial writes one record carrying the source revision, the seed, and all model scale factors, so each reported number is traceable to the configuration that produced it. The nominal flat, stair, and transition conditions are single deterministic trials and the uneven-terrain condition uses three seeds; they support reproduction and descriptive ablation ordering, not causal attribution or confidence intervals. The locked validation protocol uses paired trials, Wilson intervals for binary success, and deterministic bootstrap intervals for continuous paired differences. No physical measurement enters any table, figure, or conclusion in this paper.

VI. RESULTS

A. Flat-Ground Ablation

Table III and Fig. 4 show different trade-offs. Coordinated arc locomotion reached $0.200\,\mathrm{m\,s^{-1}}$, exceeding the $0.18\,\mathrm{m\,s^{-1}}$ command by 11.2% and travelling 71% faster than articulated walking. This is descriptively consistent with H1, but the unequal gait parameters prevent a causal claim. The same table refutes the stronger reading of H1 that we deliberately avoided: measured against the constant command, walking has the smallest tracking error ($e_v = 0.097$ against 0.253 and $0.268\,\mathrm{m\,s^{-1}}$), because rolling overshoots the command rather than following it. Rolling also consumed $3.18\times$ the absolute mechanical work and accumulated $6.23\times$ the contact slip of walking. Distal-only rotation was both slower and more expensive than walking, so the arc by itself does not explain the speed: in this model the gain is associated with coordinating posture and rotation, and it is bought with slip and work rather than with efficiency.

TABLE III

Flat nominal trial with a 0.18 m/s forward command

Controller	$\bar{v}_x$ m/s	e_v m/s	C_{mt} –	slip m
Articulated walk	0.117	**0.097**	**0.661**	**0.212**
Distal only	0.090	0.268	2.576	1.603
Coordinated arc	**0.200**	0.253	1.227	1.323

The slip is not an unexplained artifact. Equation (20) fixes the commanded stance contact speed at $v_c = 0.411\,\mathrm{m\,s^{-1}}$ for $\alpha = 1$, which is about twice the body speed the rolling condition achieves, and (21) then predicts $|v_c - \bar{v}_x|T = 1.27\,\mathrm{m}$ of integrated slip over the six-second window against $1.323\,\mathrm{m}$ measured, an error of 4.2%. For distal-only rotation the same expression predicts $1.93\,\mathrm{m}$ against $1.603\,\mathrm{m}$ measured; it over-predicts by 20% because the alignment factor A_i of (17) falls below unity when the body does not follow the contact. This arithmetic is a consistency check, not an independent prediction, because $\bar{v}_x$ is measured from the same trial and contact alignment varies over the cycle. It nevertheless shows that the fixed activity-scaled rolling command can create a large surface/body-speed mismatch. The reported slip therefore characterizes this controller setting rather than an intrinsic property of the arc geometry.

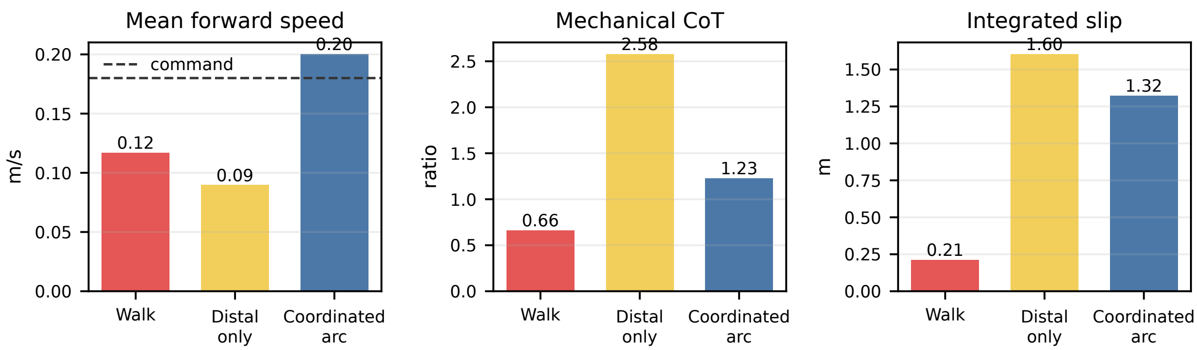


Fig. 4. Same-model flat-ground development trial (one run per controller, six-second window). Values are descriptive; error bars are intentionally absent.

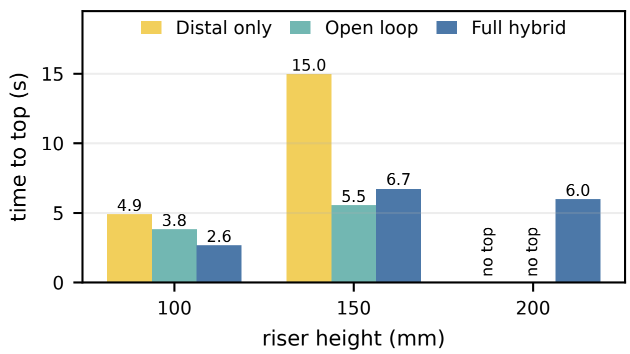


Fig. 5. Time to the upper landing for three controller restrictions over three 3-step geometries (one deterministic run each). Bars marked *no top* did not reach the landing; the binary outcomes are tabulated in Table IV.

B. Stair Coordination Ablation

Table IV compares three controller restrictions on the same open-arc model. At 100 mm every strategy succeeded. At 150 mm ($1.22r_o$), distal-only rotation reached the landing in 14.960 s with no articulation; because the riser fits inside the opening chord ($150 < l_o = 226$ mm), this outcome is consistent with edge engagement. It does not by itself isolate the opening: an actuated closed wheel can apply axle torque, so the matched closed-wheel control defined in Sec. V is required. At 200 mm ($1.63r_o$) both restricted strategies failed and only the coordinated controller reached the top, in 5.964 s with 88.18 J of work. This supports H2 only for the tested open-arc controller restrictions. Speed and safety did not coincide: at 150 mm the open-loop condition was fastest (5.546 s) but fell to $u_z = 0.459$, close to the $\cos 60°$ termination boundary, whereas the coordinated controller held $u_z = 0.755$.

C. Transitions and Uneven-Terrain Smoke Test

Walk-to-roll and roll-to-walk guards completed in 0.342 and 0.660 s. Minimum uprightness over the full transition/recovery windows was 0.991 in both directions, and neither trial produced forbidden collisions or IK failures, so H3 was not violated in either direction. On randomized uneven terrain, all nine short trials executed without the tilt

TABLE IV

Stair top-reaching outcomes (one deterministic trial)

Controller	100-mm rise top	100-mm rise s	150-mm rise top	150-mm rise s	200-mm rise top	200-mm rise s
Distal only	yes	4.886	yes	14.960	no	–
Open loop	yes	3.818	yes	5.546	no	–
Full controller	yes	2.646	yes	6.736	**yes**	5.964

termination: mean $\bar{v}_x$ over three seeds was 0.0768 ± 0.0043, 0.1025 ± 0.0356, and $0.2009 \pm 0.0126\,\mathrm{m\,s^{-1}}$ for walk, distal-only, and coordinated arc. The sample size and 2 s window do not establish robust-terrain superiority.

VII. Discussion

The experiments ask a more precise question than “can a hybrid robot walk and drive”: what behaviors can be obtained by reusing one actuated distal contact? On the open-arc model, distal rotation alone traversed the 150 mm profile and coordinated articulation was required by the tested controllers at 200 mm. On flat ground, rotation alone was slower and more expensive than walking, while the highest descriptive speed appeared when posture and rotation were commanded together. These observations motivate contact reuse—support, quasi-rolling, and candidate edge engagement from one link—but they do not yet attribute the stair result uniquely to the opening. Equation (8) explains why: a wheel driven by axle torque does not share the singular force-only bound. The matched open-arc/closed-wheel experiment is therefore the decisive test.

The costs are equally explicit. Coordinated arc locomotion produced 6.23× the slip and 3.18× the mechanical work of walking, and (20) explains the first-order mismatch: the executed controller commands an activity-scaled surface speed rather than a contact-speed feedback law. The commanded surface speed can therefore substantially exceed body speed. The present model supports a speed/cost trade-off, not an energy-efficiency claim. Rate control tied to measured body and contact velocity is a controller improvement to test separately from the morphology ablation.

Four limitations bound how far these results generalize. First, the evidence is a deterministic simulation: TPU compliance and friction, polymer infill mass, gearbox efficiency, backlash, thermal limits and battery sag are approximated, and the modeled mass remains a parameter until the assembled robot is weighed. Second, the nominal conditions are single trials, so the reported differences describe the three regimes on one model but carry no confidence intervals. Third, the restrictions are not matched against each other: the walking and rolling conditions use different cycle frequencies and stroke limits (Sec. IV-A), so the comparison is between implemented regimes rather than between equally parameterized controllers. Fourth, the edge-engagement interpretation of the 150 mm result is inferred from geometry and from the absence of articulation; it is not a direct morphology ablation. The locked paired protocol of Sec. V addresses the last three limitations in simulation, but physical model identification remains necessary. The stair supervisor is also given its geometry and performs no perception or foothold search.

VIII. Conclusion

We presented an 18-actuator hexapod whose six open arc-shaped distal links are reused for planted support, quasi-rolling contact and candidate stair-edge engagement, together with a controller that commands tripod walking and continuous arc rotation on the same joints at the same instant. In deterministic MuJoCo trials the coordinated controller reached $0.200\,\mathrm{m\,s^{-1}}$ where articulated walking reached $0.117\,\mathrm{m\,s^{-1}}$, with substantially greater work and slip; the open-arc stair restrictions traversed the 150 mm profile without articulation and the 200 mm profile only with coordinated articulation. Guarded transitions stayed upright and completed in under 0.7 s. These single-run results define hypotheses and a reproducible operating envelope, not causal morphology evidence. The accompanying matched-geometry, paired-trial protocol makes the remaining open-arc claim falsifiable. Physical speed, endurance, energy efficiency and stair reliability remain unvalidated.

References

[1] F. Tedeschi and G. Carbone, "Design of a novel leg-wheel hexapod walking robot," *Robotics*, vol. 6, no. 4, p. 40, 2017.

[2] M. Bjelonic, C. D. Bellicoso, Y. de Viragh, D. V. Sako, F. D. Tresoldi, F. Jenelten, and M. Hutter, "Keep rollin'—whole-body motion control and planning for wheeled quadrupedal robots," *IEEE Robotics and Automation Letters*, vol. 4, no. 2, pp. 2116–2123, 2019.

[3] U. Saranli, M. Buehler, and D. E. Koditschek, "RHex: A simple and highly mobile hexapod robot," *The International Journal of Robotics Research*, vol. 20, no. 7, pp. 616–631, 2001.

[4] R. D. Quinn, J. T. Offi, D. A. Kingsley, and R. E. Ritzmann, "Improved mobility through abstracted biological principles," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems*, vol. 3, 2002, pp. 2652–2657.

[5] G. Zhang, S. Ma, J. Liu, X. Zeng, L. Kong, and Y. Li, "Q-Whex: A simple and highly mobile quasi-wheeled hexapod robot," *Journal of Field Robotics*, vol. 40, no. 6, pp. 1444–1459, 2023.

[6] R. Altendorfer, N. Moore, H. Komsuoglu, M. Buehler, H. B. Brown, D. McMordie, U. Saranli, R. Full, and D. E. Koditschek, "RHex: A biologically inspired hexapod runner," *Autonomous Robots*, vol. 11, no. 3, 2001.

[7] E. Z. Moore, D. Campbell, F. Grimminger, and M. Buehler, "Reliable stair climbing in the simple hexapod RHex," in *Proc. IEEE Int. Conf. Robotics and Automation*, vol. 3, 2002, pp. 2222–2227.

[8] K. C. Galloway, G. C. Haynes, B. D. Ilhan, A. M. Johnson, R. Knopf, G. Lynch, B. Plotnick, M. White, and D. E. Koditschek, "X-RHex: A highly mobile hexapedal robot for sensorimotor tasks," University of Pennsylvania, Tech. Rep., 2010.

[9] A. Vina and A. Barrientos, "C-legged hexapod robot design guidelines based on energy analysis," *Applied Sciences*, vol. 11, no. 6, p. 2513, 2021.

[10] Y.-S. Kim, G.-P. Jung, H. Kim, K.-J. Cho, and C.-N. Chu, "Wheel transformer: A wheel-leg hybrid robot with passive transformable wheels," *IEEE Transactions on Robotics*, vol. 30, no. 6, pp. 1487–1498, 2014.

[11] S.-C. Chen, K.-J. Huang, W.-H. Chen, S.-Y. Shen, C.-H. Li, and P.-C. Lin, "Quattroped: A leg–wheel transformable robot," *IEEE/ASME Transactions on Mechatronics*, vol. 19, no. 2, 2014.

[12] W.-H. Chen, H.-S. Lin, Y.-M. Lin, and P.-C. Lin, "Turboquad: A novel leg–wheel transformable robot with smooth and fast behavioral transitions," *IEEE Transactions on Robotics*, vol. 33, no. 5, 2017.

[13] T. Yoshioka, T. Takubo, T. Arai, and K. Inoue, "Hybrid locomotion of leg–wheel ASTERISK H," *Journal of Robotics and Mechatronics*, vol. 20, no. 3, pp. 403–412, 2008.

[14] M. Bjelonic, R. Grandia, O. Harley, C. Galliard, S. Zimmermann, and M. Hutter, "Whole-body MPC and online gait sequence generation for wheeled-legged robots," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems*, 2021.

[15] M. Hutter, C. Gehring, D. Jud, A. Lauber, C. D. Bellicoso, V. Tsounis, J. Hwangbo, K. Bodie, P. Fankhauser, M. Bloesch, R. Diethelm, S. Bachmann, A. Melzer, and M. Hoepflinger, "ANYmal – a highly mobile and dynamic quadrupedal robot," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems*, 2016.

[16] V. Klemm, A. Morra, C. Salzmann, F. Tschopp, K. Bodie, L. Gulich, N. Küng, D. Mannhart, C. Pfister, M. Vierneisel, F. Weber, R. Deuber, and R. Siegwart, "Ascento: A two-wheeled jumping robot," in *Proc. IEEE Int. Conf. Robotics and Automation*, 2019.

[17] E. Todorov, T. Erez, and Y. Tassa, "MuJoCo: A physics engine for model-based control," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems*, 2012, pp. 5026–5033.